

5634083

B.Tech. DEGREE EXAMINATION, MAY 2015.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING LANGUAGES.

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. List the features of JAVA.
2. Define class.
3. Does Java support multiple inheritance? Justify the answer.
4. What do you mean by overloading?
5. Differentiate an applet and application.
6. What kind of thread is the garbage collector thread?
7. What is an inner class?
8. State the use of JDBC driver.
9. What is meant by bean customization?
10. List any two protocols used in RMI.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE questions from each Unit.

All questions carry equal marks.

UNIT I

11. What is constructor in Java? Why constructor does not have return type in Java? Explain with an example.

Or

12. What is package? What are the steps used to create a package in Java? Explain with sample code.

UNIT II

13. What is exception handling in java? Why is it used? Write a Java code to stimulate the way a stack mechanism works with exception handling, throwing and dealing with exceptions such as stack is full (if you want to add more elements into the stack) or stack is empty (if you want to pop elements from the stack).

Or

14. What is meant by stream? What are the types of streams and classes? Explain.

UNIT III

15. Name the types of layout manager. Briefly explain their operations.

Or

16. Write a program to display an application for using GUI control.

UNIT IV

17. Explain the various components of JDBC and write the steps for using JDBC to access database.

Or

18. Briefly describe about java security.

UNIT V

19. Discuss in detail about RMI.

Or

20. Develop a TCP/IP program to send the details of the salary from the server to the client. Calculate the gross pay and net pay in the client machine.

UNIT V

19. List out the steps that involved creating a new bean. Explain each step in detail. (11)

Or

20. How to create a JAR file? Explain the jar command with its all option in detail with suitable example. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. How does java support platform independency?
2. How is object destruction done in java?
3. When will a method be declared final?
4. Differentiate overloading and overriding..
5. How are the java applets used to create applications in internet?

6. Which class is the top of the AWT event hierarchy?

7. Which Java, util classes and interfaces support event handling?

8. List the difference between Java inner class and anonymous class.

9. What is the use of Java RMI?

10. What is meant by socket programming?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. (a) Write short notes on features of Java and Java Virtual Machine. (6)

(b) What is the use of Constructors and how it is implemented in JAVA? (5)

Or

12. Define package in JAVA and explain how to add a class or interface to a package? (11)

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UNIT II

13. What is multithreading? What are the methods available, in java for inter - thread communication? Discuss with an example. (11)

Or

14. Define interface and explain how it is used to implement multiple inheritance. (11)

UNIT III

15. Describe the types of layout management and swing components briefly. (11)

Or

16. Elaborately discuss the components of graphics programming. Illustrate the basic of event handling. (11)

UNIT IV

17. Explain the concept of Object cloning and inner classes with example. (11)

Or

18. State the motivations of generic programming. Explain the generic classes and methods with example. (11)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is a ByteCode?
2. What is an Interface?
3. What is operator Overloading?
4. What is Multithreading?
5. Give the purpose of containers.
6. Give the significance of swing class.
7. What is JDBC?

8. What are Utility packages?
9. What is an InetAddress?
10. What is socket?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Write a Java Program to perform Matrix Multiplication.

Or

12. Discuss about the control structures in Java.

UNIT II

13. Explain how operator overloading can be used for arithmetic in Complex numbers.

Or

14. Explain Exception handling with an example.

UNIT III

15. Write a java program to display an image using applets.

Or

16. Discuss the importance of Garbage Collection and methodology in Java for the same.

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UNIT IV

17. Discuss with an example the advantages of using Input Output Packages.

Or

18. Explain the steps for using the JDBC with an example.

UNIT V

19. Explain the steps in using the Bean Development Kit.

Or

20. Explain the facility in Java for Remote Method Invocation.

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16. (a) Explain about the model view controller design pattern. (5)
 (b) What is the function of layout manager? Describe in detail about the different layout in Java GUI. (6)

UNIT IV

17. Write a Java program to find total marks of student and updates the table and displays the students information using JDBC. (Student name, sub1, sub2, sub3, total). (11)

Or

18. Write a simple generic version of method "is equal to" that compares its two arguments with the equals method and returns true if they are equal and false otherwise. Use this generic method in a program that calls "is equals to" with a variety of built-in types such as objector integer. (11)

UNIT V

19. (a) What is Java Bean? How to make it a persistent one? Why enterprise Java Beans are used? (5)
 (b) Write RMI client server to multiply two integers and display the result in the client. (6)

Or

20. (a) What is meant by session tracking? How session tracking done with API? (5)
 (b) Describe the Java bean component architecture. (6)

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B.Tech. DEGREE EXAMINATION, DECEMBER 2016.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Compare Garbage collection and finalize method?
2. What is a String in Java?
3. What are object wrappers? Give example.
4. How to create custom exceptions?
5. Give the syntax of drawImage () and copyArea() methods.
6. What are the methods under WindowListener Interface?
7. What is local inner class and anonymous inner class? Give their advantages.

8. Name some Java API Packages.

9. What types of application prefer UDP rather than TCP?

10. How to set-up TCP connection?

PART B — (5 × 11 = 55 marks)

Answer ALL the Questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Develop an employee class which implements the comparable and Cloneable interfaces. Implement the Sorting of persons (based on name in alphabetical). Also implement the shallow copy (for name and age) and deep Copy (for Date of Joining). (11)

Or

12. Write a Java program to maintain the books details like BookId, accession number, book name, author, publication in books Package and keep the journal details such as journal Id, journal name in journal Package in main class. Use these two Packages details for staff and student classes and display the books and journals information as requested by the user. (11)

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UNIT II

13. (a) How to extend the thread class? Give an example. (6)

(b) Discuss the ways in which inheritance promotes software reuse, saves time during program development and helping in preventing errors. Give an example. (5)

Or

14. (a) Write a program in Java to explain the creation of user-defined exception and explain how to handle it. (6)

(b) Explain about the thread synchronization with an example. (5)

UNIT III

15. Write a Java program to handle the mouse events for the following

- Mouse Pressed - Print Mouse is pressed with current x and y position
- Mouse release - Print Mouse is released with current x and y position
- Mouse dragged - Print Mouse is released with current x and y position
- Mouse moved - Print Mouse is pressed with current x and y position.

Or

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UNIT IV

17. What is need for the Thread? How you create multiple threads? How do they work together when multiple works are under execution? Explain with an example. (11)

Or

18. Write down the socket programming so that the client and server at different locations are able to communicate to each other. (11)

UNIT V

19. (a) What is byte code? What are the significances of it? (3)
(b) Write down a Java code to add 'n' elements retrieved from the data base and store in the vector collection class objects. (8)

Or

20. (a) What are the advantages of object serialization? Justify whether the property of serialization to be kept at the very base class level (generalization) or at very derived class (specialization). Comment on your statements. (6)
(b) What are garbage collections? What are the features available in Java language for this purpose? Explain briefly. (5)

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016,

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING LANGUAGES

(For 2009-2012 Batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Compare Class and Object.
2. List out any four features of object oriented programming language.
3. List out the operators which cannot be overloaded.
4. Compare and give an example for compile and runtime polymorphism.
5. How do you create an array of objects in Java language?
6. List out the types of inheritance in Java language.

8. What is the purpose of the finally clause of a try-catch-finally statement?

9. What are the states of a thread?

10. What are the types of Java beans?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each unit.

All questions carry equal marks.

UNIT I

11. Write a code in java to generate single calculator using classes and accepting the two integers and operator with all methods to input, display, add, subtract, product and division.

Or

12. Explain method overriding with example program.

UNIT II

13. What is dynamic binding? Explain with example.

Or

14. Explain the concept of inheritance and its types.

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UNIT III

15. Write a code in java which take 10 numbers store it in a combo box and display the greatest one.

Or

16. Write a program to develop a login window.

UNIT IV

17. Explain the steps to set up Java Database Connectivity.

Or

18. With a sample program explain How Java handles integer overflows and underflows?

UNIT V

19. Explain about get and set methods in the Datagram Packet class.

Or

20. Create a client / server application using RMI in Java to perform remote calculation.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

OBJECT ORIENTED PROGRAMMING

(2013 - 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the pillars of OOPS concept?
2. What is an access specifier?
3. Define Throw.
4. What is a super constructor?
5. Define Auto Boxing.
6. What are the uses of references in functions?
7. What is dynamic binding?

8. What are wrapper classes?
9. How to create a TCP/IP socket?
10. Define persistence.

PART B — (5 × 11 = 55 marks)

Answer ALL the Questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Write a java program that contains an array of employees (name, salary, ID) - and a static variable country. List the number of employees along with the country of the employees. (6)
- (b) Explain about packages with examples. (5)

Or

12. Write a program that accomplishes each of the task:
 - (a) Write a method that calculates the integer part of the quotient when integer a is divided by integer b. (3)
 - (b) Write a method that calculates the integer remainder when integer a is divided by integer b. (4)

- (c) -Write a method that display digits that receives integer between 1 and 999 and display it as a sequence of digits separating each pair of digits by two spaces. (4)

UNIT II

13. Explain Multi. threading with suitable example. Write a simple java program to explain the concept.

Or

14. (a) What is inheritance? Write a program to inheriting a class. (6)
- (b) What is exception? How to throw an exception? Give an example. (6)

UNIT III

15. (a) Explain the AWT Event handling in detail. (10)
- (b) Give the methods available in graphics COLOR. (10)

Or

16. With example explain the methods in Model view controller. (10)

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B.Tech- DEGREE EXAMINATION, NOVEMBER 2017

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an object? Give an example.
2. What are the uses of destructor?
3. Define a thread.
4. Name the standard streams available in JAVA?
5. Define AWT.
6. List out the listener methods for implementing the KeyListener interface.

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7. Define a generic class with an example.
8. What is an inner class? Give an example.
9. Define URL.
10. What is a bean persistence property?

PART B — (5 × 11 = 55 marks)

Answer ALL questions. One question from each Unit.

All questions carry equal marks.

UNIT I

11. Elaborate on the various object oriented concepts with necessary illustrations.

Or

12. (a) What is constructor in JAVA? Why constructor does not have return type in JAVA? Explain with an example. (5)

- (b) Write a JAVA program with self explanatory comments which implements the concept of interface. (6)

UNIT II

13. Explain with an example how multiple inheritance is achieved in JAVA.

Or

14. Discuss about the JAVA error handling mechanism and what is the difference between "unchecked exception and checked exception"?

UNIT III

15. List and briefly discuss about the various swing components in JAVA.

Or

16. Write an applet program that mimics a calculator which takes two real numbers and performs any four arithmetic operations.

UNIT IV

17. Using generic class write a program to perform the following operations on an array.

- (a) Add an element in the beginning middle/end. (6)

Or

18. Explain in detail about the database connectivity and security enforcing mechanism in JAVA with an example.

UNIT V

19. Discuss briefly about RMI.

Or

20. Define JAVA bean and list out the steps for creating a new bean.

3. Under what situation the keywords "try", "catch" is used?
4. List out the difference between overriding and overloading in java.
5. Compare AWT and Swing (Any two points).
6. What is the purpose of garbage collection in java and when it is used?
7. List out any two key security features provided by java programming language.
8. What are the benefits of generics in java?
9. Define Datagram.
10. List out any two advantages of Java Beans.

PART B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. What is interface? What is the use of Interface in java? How to create it and access it? Explain with example.
12. Define Constructors. What is the use of constructors? Explain various types of constructors with example.

Or

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UNIT II

13. What do you understand by inheritance? Explain each type of inheritance with the help of example and diagram.
14. Write a program to copy content of one file to another using character stream class.

Or

UNIT III

15. What is Event class? Write a program for mouse event handler.

Or

16. Design Simple calculator using java, Swing components.

UNIT IV

17. Brief the steps involved in connecting to the database using java. Write an example program to Connect Java Application with mysql database.

Or

18. Explain briefly the Input Output Stream class with example.

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UNIT IV

17. Write short notes on

- (a) Utility packages. (5)
- (b) Database Connectivity. (6)

Or

- 18. (a) Discuss the Object and inner classes with example. (7)
- (b) What is a static inner class? Explain with example. (4)

UNIT V

19. Write a program in java for the following

Create a server socket that accepts incoming connections. Let a client send its socket information to the server. If the client send requests more than 2 times, through the same socket the server should send a response saying "GOODBYE" and should not respond for any further request.

Or

- 20. (a) Write a java program to display the IP address of the system in which the java program is running. (6)
- (b) Draw the architecture of Java RMI. Write an RMI application to display a message from the remote server. (5)

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Fourth Semester

Computer Science and Engineering / Information Technology

OBJECT ORIENTED PROGRAMMING

(2013 - 2014 onwards)

Time: Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Mention the features of Java.
2. What do you mean by reflection?
3. Define static binding and dynamic binding.
4. What is the need of finally clause?
5. Swing component are not thread safe, why?
6. What is the relationship between event sources and listeners?
7. Define generic classes.

8. What is the difference between system. out. print and system. out. println in Java?

9. Define Package.

10. Define API.

PART B — (5 × 11 = 55 marks)

Answer ALL Questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Write a Java program to illustrate the difference between an instance variable and a class variable.

Or

12. Define a class string with appropriate constructors, destructor and overloaded +, =, and == operators use them in a main driver program.

UNIT II

13. What are the different forms of inheritance? Give example for them.

Or

14. Write a C++ program containing a possible exception. Use a try block to throw it and a catch block to handle it properly.

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UNIT III

15. Explain Applet life cycle and also explain how to provide parameters through HTML with example.

Or

16. Explain AWT package with a suitable example.

UNIT IV

17. (a) Explain package in Java. List out all packages with short description.

(b) Differentiate Method Overloading and Method Overriding.

Or

18. Briefly explain Generics in Java with demo program.

UNIT V

19. Explain RMI and steps to write the RMI program.

Or

20. (a) How to print total numbers of tables and views of a database using JDBC?

(b) How to store and retrieve images from Oracle database using JDBC?

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9. Write a java program to implement TCP client and server. (11)

Or

10. Develop a java program to implement RMI client and server. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013 - 14 onwards batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Why java language is called as platform independent language? How cross platform capability is achieved by JAVA?
2. What is interface? List down the applications of interfaces.
3. What is function overloading? Why it is necessary in object oriented programming?
4. What is multithreading? What are the advantages of multithreading?
5. What is event driven programming?

6. Write a Swing program to add one button and make it visible to JFrame object.

7. What is Object Serialization?

8. Differentiate inner class and nested class.

9. What are properties of java bean class?

10. What is RMI?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

11. (a) Explain in detail about various control structures used in java. (6)

(b) Write a java program to store and retrieve student mark details using classes and objects. (5)

Or

12. (a) Explain in detail about constructors, destructors and interfaces in java. (6)

(b) Create one interface with four methods add, subtract, multiply and divide. Write a java program to implement this interface. (5)

13. Explain in detail about overloading and inheritance with examples. (11)

Or

14. Explain in detail about java exception handling with examples. (11)

15. Develop a java applet program with two text boxes and one button. When user enters text in text boxes and click the button, check and display whether both texts are equal or not. (11)

Or

16. Develop a JFrame for basic calculator application using java swing. (11)

17. With example program discuss in detail about java array, arraylist, calendar and date classes. (11)

Or

18. Develop a JDBC application to connect with employee table that contains four columns ID, NAME, SALARY and PHONE NO. Display all employee details and double the salary of the employee with ID=123. (11)

UNIT V

19. Write an Java programme for Socket Client/Server communication.

Or

20. Write a Java programme for displaying source code of a webpage by URL Connection class.

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013-14 onwards Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Write any two features of Java.

2. What is the output of the following Java program?

```
class Test
{
    public static void main (String args[])
    {
        System.out.println(10 + 20 + "Javatpoint");
        System.out.println("Javatpoint" + 10 + 20);
    }
}
```


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B. Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Fourth Semester

CSE

OBJECT ORIENTED PROGRAMMING

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A.— (10 × 2 = 20 marks)

Answer ALL the questions.

1. List out the features of Java.
2. Mention how interfaces differ from abstract class.
3. Define inheritance and its types.
4. Deduce the meaning for the keywords: Final, Finally, Finalize.
5. What is event handling?
6. Write an applet code to draw smiley.
7. What are collections in Java?

8. Define anonymous inner class.
9. List some advantages of Java Beans.
10. Define RMI.

PART B — (5 × 11 = 55 marks)

Answer ONE question from each Unit.

UNIT – I

11. Explain the basic concepts of object oriented programming and mention its limitations.

Or

12. Explain Packages in detail with a program.

UNIT – II

13. Elaborate the concept of Inheritance and its types.

Or

14. Explain the concept of Multithreading with suitable example program.

UNIT – III

15. Develop a Student information System with necessary AWT components and trigger and handle the events.

Or

16. List out the advantages of Swing in Java. Develop an application using swing components.

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UNIT – IV

17. Write short notes on Input/Output streams in java.

Or

18. Elucidate inner classes with its types with example program for each.

UNIT – V

19. Elucidate about java bean in detail with an example program.

Or

20. Explain the concept of Socket with a suitable client/server code.

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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Fourth Semester

Computer Science and Engineering

OBJECT ORIENTED PROGRAMMING

(2013-14 Regulation)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define the term Class.
2. What is called Constructor?
3. Define File.
4. Write about the term Stream.
5. Explain about layout.
6. Write a note on Garbage collection.
7. What is called Generics?

8. Explain the term Utility Packages.

9. Expand : TCP/IP and HTTP.

10. What is called URL Connection?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain about the term Operators.

(b) Give a note on Destructor.

Or

12. (a) Discuss the term Control Structures.

(b) Write a note on Interfaces.

UNIT II

13. (a) Describe about the term Overloading.

(b) Write a note on Exception Handling.

Or

14. (a) Explain the concept Multithreading.

(b) Discuss about the term Inheritance.

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UNIT III

15. (a) What are called GUI components? Explain.

(b) Discuss the fundamentals of Swing.

Or

16. (a) Explain about the AWT package.

(b) Write a note on Swing class.

UNIT IV

17. (a) Explain about the Utility Packages.

(b) List the concepts in Inner Class.

Or

18. (a) Write a note on Input-output Packages.

(b) Discuss the term Java Security.

UNIT V

19. (a) Explain about the Application Builder tool.

(b) Write a note on TCP/IP Client sockets.

Or

20. (a) What is called JAR files? Explain.

(b) Write a note on RMI.

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B.Tech. DEGREE EXAMINATION, MARCH 2021.

Fourth Semester

Computer Science Engineering

OBJECT ORIENTED PROGRAMMING

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks

1. List out the features of Java.
2. Define destructor.
3. What is function overloading?
4. What is abstract class?
5. List out the GUI components.
6. What are streams in Java?
7. Define an interface.
8. How swing classes are used in Java?
9. Write notes on datagrams.
10. What is BDK?

PART B — (5 × 11 = 55 marks)

Answer all the questions. All questions carry equal marks

UNIT I

11. Elaborate the operators used in Java programming.

Or

12. Discuss the concept of inheritance with supporting example.

UNIT II

13. Describe the exception handling mechanism using keywords that support the same.

Or

14. Detail about multithreading in Java.

UNIT III

15. Explain the life cycle of an Applet. Write a Applet program to display an image file.

Or

16. Describe the basics of event handling.

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UNIT IV

17. Discuss about Collection classes and Collection interfaces.

Or

18. How database is connected to a Java programs? Illustrate with an example.

UNIT V

19. Elaborate on TCP/IP and write a client side socket programming to count the words to read phrases that start with '\$' sign.

Or

20. Discuss on RMI in Java and write down the steps to do it programmatically.

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5	4	3	2	1	Time
5	4	3	2	1	Time